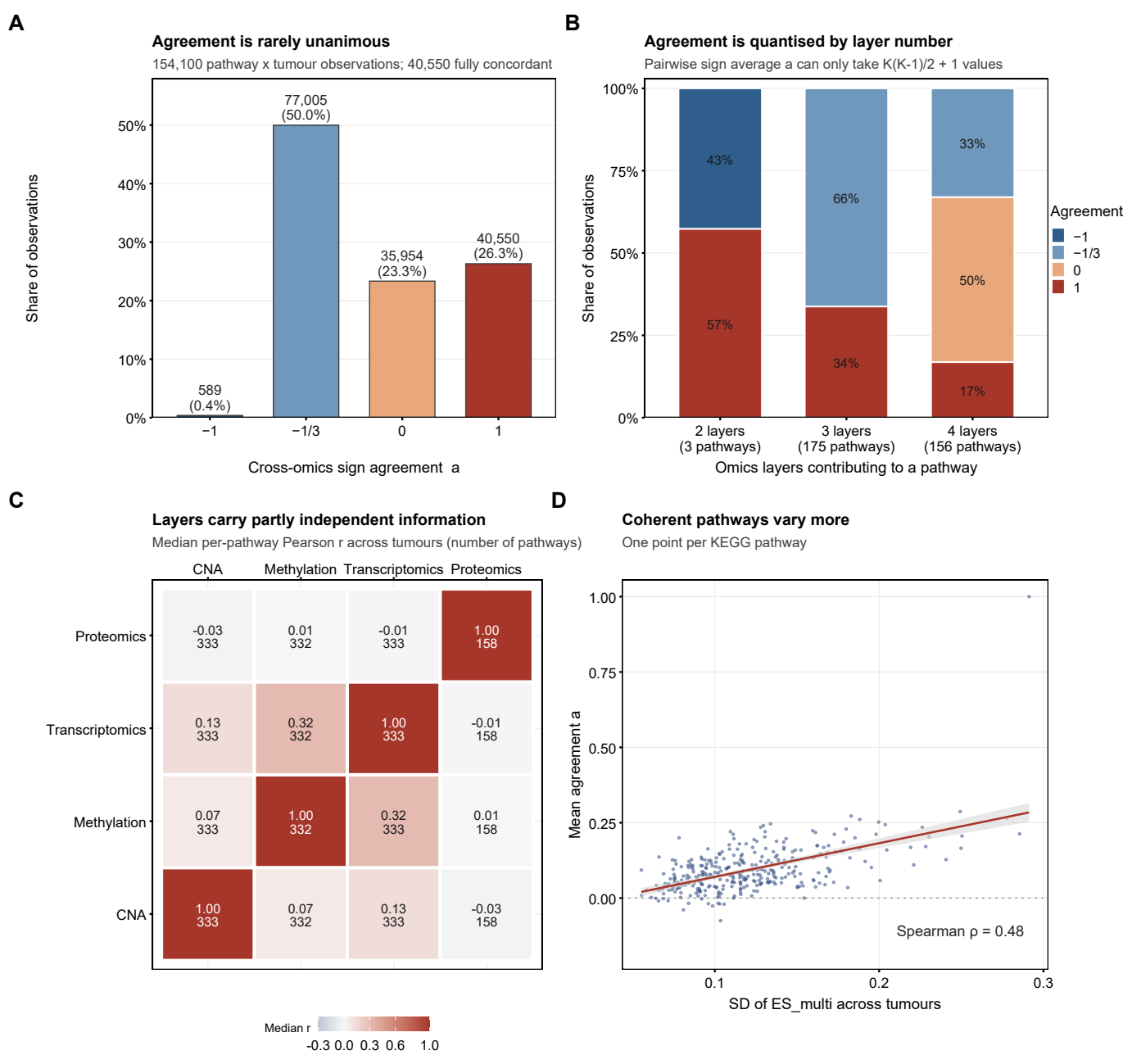

A
Agreement is rarely unanimous
154,100 pathway x tumour observations; 40,550 fully concordant
Share of observations
77,005
(50.0%)
35,954
(23.3%)
40,550
(26.3%)
589
(0.4%)
50%
40%
30%
20%
10%
0%
−1
−1/3
0
1
Cross-omics sign agreement a
B
Agreement is quantised by layer number
Pairwise sign average a can only take K(K-1)/2 + 1 values
Share of observations
100%
75%
50%
25%
0%
43%
57%
66%
34%
33%
50%
17%
2 layers
(3 pathways)
3 layers
(175 pathways)
4 layers
(156 pathways)
Omics layers contributing to a pathway
Agreement
−1
−1/3
0
1
C
Layers carry partly independent information
Median per-pathway Pearson r across tumours (number of pathways)
CNA
Methylation
Transcriptomics
Proteomics
Proteomics
Transcriptomics
Methylation
CNA
-0.03
333
0.01
332
-0.01
333
1.00
158
0.13
333
0.32
332
1.00
333
-0.01
158
0.07
333
1.00
332
0.32
333
0.01
158
1.00
333
0.07
332
0.13
333
-0.03
158
Median r
-0.3 0.0 0.3 0.6 1.0
D
Coherent pathways vary more
One point per KEGG pathway
Mean agreement a
1.00
0.75
0.50
0.25
0.00
Spearman ρ = 0.48
0.1
0.2
0.3
SD of ES_multi across tumours